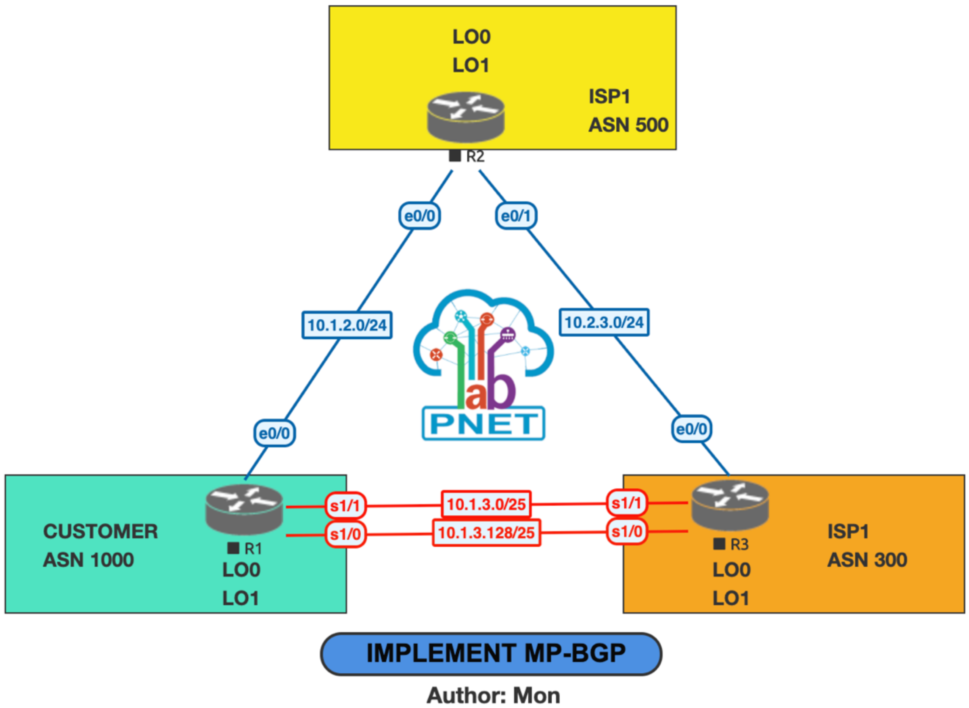
IMPLEMENT MP-BGP

Lab Topology:



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Addressing Table:

Device	Interface	Ipv4 address	Ipv6 address	Ipv6 Link-local
R1	e0/0	10.1.2.1/24	2001:db8:acad:1012::1/64	fe80::1:1
	S1/0	10.1.3.1/25	2001:db8:acad:1013::1/64	fe80::1:2
	S1/1	10.1.3.129/25	2001:db8:acad:1014::1/64	fe80::1:3
	Loopback0	192.168.1.1/27	2001:db8:acad:1000::1/64	fe80::1:4
	Loopback1	192.168.1.65/26	2001:db8:acad:1001::1/64	fe80::1:5
R2	e0/0	10.1.2.2/24	2001:db8:acad:1012::2/64	fe80::2:1
	e0/1	10.2.3.2/24	2001:db8:acad:1023::2/64	fe80::2:2
	Loopback0	192.168.2.1/27	2001:db8:acad:2000::1/64	fe80::2:3
	Loopback1	192.168.2.65/26	2001:db8:acad:2001::1/64	fe80::2:4
R3	e0/0	10.2.3.3/24	2001:db8:acad:1023::3/64	fe80::3:1
	S1/0	10.1.3.3/25	2001:db8:acad:1013::3/64	fe80::3:2
	S1/1	10.1.3.130/25	2001:db8:acad:1014::3/64	fe80::3:3
	Loopback0	192.168.3.1/27	2001:db8:acad:3000::1/64	fe80::3:4
	Loopback1	192.168.3.65/26	2001:db8:acad:3001::1/64	fe80::3:5

Objectives:

The objective of lab exercise is for you to learn and understand step-by-step configure MP-BGP.

Task:

1. **Build the Network and Configure Basic Device Settings and Interface Addressing**
2. **Configure MP-BGP on all Routers**
3. **Verify MP-BGP**
4. **Configure and Verify IPv6 Summarization**

Solution:

Task 1: Build the Network and Configure Basic Device Settings

Router R1

```
hostname R1
no ip domain lookup
line con 0
 logging sync
 exec-time 0 0
 exit
interface Loopback0
 ip address 192.168.1.1 255.255.255.224
 ipv6 address FE80::1:4 link-local
 ipv6 address 2001:DB8:ACAD:1000::1/64
 no shut
interface Loopback1
 ip address 192.168.1.65 255.255.255.192
 ipv6 address FE80::1:5 link-local
 ipv6 address 2001:DB8:ACAD:1001::1/64
 no shut
interface e0/0
 ip address 10.1.2.1 255.255.255.0
 ipv6 address FE80::1:1 link-local
 ipv6 address 2001:DB8:ACAD:1012::1/64
 no shut
interface s1/0
 ip address 10.1.3.1 255.255.255.128
 ipv6 address FE80::1:2 link-local
```

```
 ipv6 address 2001:DB8:ACAD:1013::1/64
 no shut
interface s1/1
 ip address 10.1.3.129 255.255.255.128
 ipv6 address FE80::1:3 link-local
 ipv6 address 2001:DB8:ACAD:1014::1/64
 no shut
```

Router R2

```
hostname R2
no ip domain lookup
line con 0
 logging sync
 exec-time 0 0
 exit
interface Loopback0
 ip address 192.168.2.1 255.255.255.224
 ipv6 address FE80::2:3 link-local
 ipv6 address 2001:DB8:ACAD:2000::1/64
 no shut
interface Loopback1
 ip address 192.168.2.65 255.255.255.192
 ipv6 address FE80::2:4 link-local
 ipv6 address 2001:DB8:ACAD:2001::1/64
 no shut
interface e0/0
 ip address 10.1.2.2 255.255.255.0
 ipv6 address FE80::2:1 link-local
 ipv6 address 2001:DB8:ACAD:1012::2/64
 no shut
interface e0/1
 ip address 10.2.3.2 255.255.255.0
 ipv6 address FE80::2:2 link-local
 ipv6 address 2001:DB8:ACAD:1023::2/64
 no shut
```

Router R3

```
hostname R3
no ip domain lookup
line con 0
 logging sync
 exec-time 0 0
 exit
interface Loopback0
 ip address 192.168.3.1 255.255.255.224
 ipv6 address FE80::3:4 link-local
 ipv6 address 2001:DB8:ACAD:3000::1/64
 no shut
interface Loopback1
 ip address 192.168.3.65 255.255.255.192
 ipv6 address FE80::3:5 link-local
 ipv6 address 2001:DB8:ACAD:3001::1/64
 no shut
interface e0/0
 ip address 10.2.3.3 255.255.255.0
 ipv6 address FE80::3:1 link-local
 ipv6 address 2001:DB8:ACAD:1023::3/64
 no shut
interface s1/0
 ip address 10.1.3.3 255.255.255.128
 ipv6 address FE80::3:2 link-local
 ipv6 address 2001:DB8:ACAD:1013::3/64
 no shut
interface s1/1
 ip address 10.1.3.130 255.255.255.128
 ipv6 address FE80::3:3 link-local
 ipv6 address 2001:DB8:ACAD:1014::3/64
 no shut
```

Task 2: Configure MP-BGP on all Routers

Step 1: Implement eBGP and neighbor relationships on R1 for IPv4 and IPv6.

a. Enable IPv6 routing:

```
R1(config)# ipv6 unicast-routing
```

b. Enter BGP configuration mode from global configuration mode, specifying AS 1000 and configure the router ID.

```
R1(config)# router bgp 1000
R1(config-router)# bgp router-id 1.1.1.1
```

c. Based on the topology diagram, configure all the designated IPv4 neighbors for R1.

```
R1(config-router)# neighbor 10.1.2.2 remote-as 500
R1(config-router)# neighbor 10.1.3.3 remote-as 300
R1(config-router)# neighbor 10.1.3.130 remote-as 300
```

d. Based on the topology diagram, configure all the designated IPv6 neighbors for R1.

```
R1(config-router)# neighbor 2001:db8:acad:1012::2 remote-as 500
R1(config-router)# neighbor 2001:db8:acad:1013::3 remote-as 300
R1(config-router)# neighbor 2001:db8:acad:1014::3 remote-as 300
```

e. Enter address family configuration mode for IPv4 and activate each of the IPv4 neighbors.

```
R1(config-router)# address-family ipv4 unicast
R1(config-router-af)# neighbor 10.1.2.2 activate
R1(config-router-af)# neighbor 10.1.3.3 activate
R1(config-router-af)# neighbor 10.1.3.130 activate
R1(config-router-af)# exit
```

f. Enter address family configuration mode for IPv6 and activate each of the IPv6 neighbors.

```
R1(config-router)# address-family ipv6 unicast
R1(config-router-af)# neighbor 2001:db8:acad:1012::2 activate
R1(config-router-af)# neighbor 2001:db8:acad:1013::3 activate
R1(config-router-af)# neighbor 2001:db8:acad:1014::3 activate
R1(config-router-af)# exit
```

Step 2: Implement eBGP and neighbor relationships on R2 for IPv4 and IPv6.

a. Enable IPv6 routing.window

```
R2(config)# ipv6 unicast-routing
```

b. Enter BGP configuration mode from global configuration mode, specifying AS 500 and configure the router ID.

```
R2(config)# router bgp 500
R2(config-router)# bgp router-id 2.2.2.2
```

c. Based on the topology diagram, configure all the designated IPv4 neighbors for R1.

```
R2(config-router)# neighbor 10.1.2.1 remote-as 1000
R2(config-router)# neighbor 10.2.3.3 remote-as 300
```

d. Based on the topology diagram, configure all the designated IPv6 neighbors for R1.

```
R2(config-router)# neighbor 2001:db8:acad:1012::1 remote-as 1000
R2(config-router)# neighbor 2001:db8:acad:1023::3 remote-as 300
```

e. Enter address family configuration mode for IPv4 and activate each of the IPv4 neighbors.

```
R2(config-router)# address-family ipv4 unicast
R2(config-router-af)# neighbor 10.1.2.1 activate
R2(config-router-af)# neighbor 10.2.3.3 activate
R2(config-router-af)# exit
```

f. Enter address family configuration mode for IPv6 and activate each of the IPv6 neighbors.

```
R2(config-router)# address-family ipv6 unicast
R2(config-router-af)# neighbor 2001:db8:acad:1012::1 activate
R2(config-router-af)# neighbor 2001:db8:acad:1023::3 activate
R2(config-router-af)# exit
```

Step 3: Implement eBGP and neighbor relationships on R3 for IPv4 and IPv6.

a. Enable IPv6 routing.

```
R3(config)# ipv6 unicast-routing
```

b. Enter BGP configuration mode from global configuration mode, specifying AS 300 and configure the router ID.

```
R3(config)# router bgp 300
R3(config-router)# bgp router-id 3.3.3.3
```

c. Based on the topology diagram, configure all the designated IPv4 neighbors for R1.

```
R3(config-router)# neighbor 10.2.3.2 remote-as 500
R3(config-router)# neighbor 10.1.3.1 remote-as 1000
R3(config-router)# neighbor 10.1.3.129 remote-as 1000
```

d. Based on the topology diagram, configure all the designated IPv6 neighbors for R1.

```
R3(config-router)# neighbor 2001:db8:acad:1023::2 remote-as 500
R3(config-router)# neighbor 2001:db8:acad:1013::1 remote-as 1000
R3(config-router)# neighbor 2001:db8:acad:1014::1 remote-as 1000
```

e. Enter address family configuration mode for IPv4 and activate each of the IPv4 neighbors.

```
R3(config-router)# address-family ipv4 unicast
R3(config-router-af)# neighbor 10.1.3.1 activate
R3(config-router-af)# neighbor 10.1.3.129 activate
R3(config-router-af)# neighbor 10.2.3.2 activate
R3(config-router-af)# exit
```

f. Enter address family configuration mode for IPv6 and activate each of the IPv6 neighbors.

```
R3(config-router)# address-family ipv6 unicast
R3(config-router-af)# neighbor 2001:db8:acad:1023::2 activate
R3(config-router-af)# neighbor 2001:db8:acad:1013::1 activate
R3(config-router-af)# neighbor 2001:db8:acad:1014::1 activate
R3(config-router-af)# exit
```

Step 4: Advertise IPv4 and IPv6 prefixes on R1.

a. Enter address family configuration mode for IPv4 and advertise the IPv4 prefixes.

```
R1(config-router)# address-family ipv4 unicast
R1(config-router-af)# network 192.168.1.0 mask 255.255.255.224
R1(config-router-af)# network 192.168.1.64 mask 255.255.255.192
R1(config-router-af)# exit
```

b. Enter address family configuration mode for IPv6 and advertise the IPv6 prefixes.

```
R1(config-router)# address-family ipv6 unicast
R1(config-router-af)# network 2001:db8:acad:1000::/64
R1(config-router-af)# network 2001:db8:acad:1001::/64
R1(config-router-af)# exit
```

Step 5: Advertise IPv4 and IPv6 prefixes on R2.

a. Enter address family configuration mode for IPv4 and advertise the IPv4 prefixes.

```
R2(config-router)# address-family ipv4 unicast
R2(config-router-af)# network 192.168.2.0 mask 255.255.255.224
R2(config-router-af)# network 192.168.2.64 mask 255.255.255.192
R2(config-router-af)# exit
```

b. Enter address family configuration mode for IPv6 and advertise the IPv6 prefixes.

```
R2(config-router)# address-family ipv6 unicast
R2(config-router-af)# network 2001:db8:acad:2000::/64
R2(config-router-af)# network 2001:db8:acad:2001::/64
R2(config-router-af)# exit
```

Step 6: Advertise IPv4 and IPv6 prefixes on R3.

a. Enter address family configuration mode for IPv4 and advertise the IPv4 prefixes.

```
R3(config-router)# address-family ipv4 unicast
R3(config-router-af)# network 192.168.3.0 mask 255.255.255.224
R3(config-router-af)# network 192.168.3.64 mask 255.255.255.192
R3(config-router-af)# exit
```

b. Enter address family configuration mode for IPv6 and advertise the IPv6 prefixes.

```
R3(config-router)# address-family ipv6 unicast
```

```
R3(config-router-af)# network 2001:db8:acad:3000::/64
R3(config-router-af)# network 2001:db8:acad:3001::/64
R3(config-router-af)# exit
```

Task 3: **Verify MP-BGP**

Step 1: Display summary neighbor adjacency information.

```
R2#show bgp ipv4 unicast summary
BGP router identifier 2.2.2.2, local AS number 500
BGP table version is 7, main routing table version 7
6 network entries using 840 bytes of memory
10 path entries using 800 bytes of memory
5/3 BGP path/bestpath attribute entries using 720 bytes of memory
4 BGP AS-PATH entries using 96 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 2456 total bytes of memory
BGP activity 12/0 prefixes, 20/0 paths, scan interval 60 secs

Neighbor    V    AS   MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
10.1.2.1    4    1000  17       17      7       0   0   00:11:36    4
10.2.3.3    4    300   16       15      7       0   0   00:09:30    4
```

```
R2#show bgp ipv6 unicast summary
BGP router identifier 2.2.2.2, local AS number 500
BGP table version is 7, main routing table version 7
6 network entries using 984 bytes of memory
10 path entries using 1040 bytes of memory
5/3 BGP path/bestpath attribute entries using 720 bytes of memory
4 BGP AS-PATH entries using 96 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 2840 total bytes of memory
```

```
BGP activity 12/0 prefixes, 20/0 paths, scan interval 60 secs

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down
State/PfxRcd
2001:DB8:ACAD:1012::1
                4         1000      18      18        7    0    0 00:11:49        4
2001:DB8:ACAD:1023::3
          4         300      15      15        7    0    0 00:09:15        4
```

Step 2: Verify BGP tables for IPv4 and IPv6.

```
R2#show bgp ipv4 unicast
BGP table version is 7, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *   192.168.1.0/27   10.2.3.3                           0 300 1000 i
 *>                   10.1.2.1                 0          0 1000 i
 *   192.168.1.64/26  10.2.3.3                           0 300 1000 i
 *>                   10.1.2.1                 0          0 1000 i
 *>  192.168.2.0/27   0.0.0.0                  0       32768 i
 *>  192.168.2.64/26  0.0.0.0                  0       32768 i
 *   192.168.3.0/27   10.1.2.1                           0 1000 300 i
 *>                   10.2.3.3                 0          0 300 i
 *   192.168.3.64/26  10.1.2.1                           0 1000 300 i
*>                    10.2.3.3                 0          0 300 i
```

R2#show bgp ipv6 unicast

```
BGP table version is 7, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,    r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter, x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

          Network                  Next Hop                Metric LocPrf Weight Path
*  2001:DB8:ACAD:1000::/64  2001:DB8:ACAD:1023::3      0     300    1000    i
*>                          2001:DB8:ACAD:1012::1      0       0    1000    i
*  2001:DB8:ACAD:1001::/64  2001:DB8:ACAD:1023::3      0     300    1000    i
*>                          2001:DB8:ACAD:1012::1      0       0    1000    i
*> 2001:DB8:ACAD:2000::/64  ::                         0              32768  i
*> 2001:DB8:ACAD:2001::/64  ::                         0              32768  i
```

Step 3: Viewing explicit routes and path attributes

```
R2# show bgp ipv4 unicast 192.168.1.0 255.255.255.224
BGP routing table entry for 192.168.1.0/27, version 2
Paths: (2 available, best #2, table default)
Advertised to update-groups: 1
Refresh Epoch 1
300 1000
10.2.3.3 from 10.2.3.3 (3.3.3.3) Origin IGP, localpref 100, valid, external
rx pathid: 0, tx pathid: 0
Refresh Epoch 1
1000
10.1.2.1 from 10.1.2.1 (1.1.1.1)
Origin IGP, metric 0, localpref 100, valid, external, best rx pathid: 0, tx pathid: 0x0
```

```
R2# show bgp ipv6 unicast 2001:db8:acad:1000::/64
BGP routing table entry for 2001:DB8:ACAD:1000::/64, version 2
Paths: (2 available, best #2, table default)
  Flag: 0x100
  Advertised to update-groups: 1
Refresh Epoch 1
```

```
300 1000
2001:DB8:ACAD:1023::3 (FE80::3:1) from 2001:DB8:ACAD:1023::3 (3.3.3.3) Origin IGP,
localpref 100, valid, external
rx pathid: 0, tx pathid: 0
Refresh Epoch 1
1000
2001:DB8:ACAD:1012::1 (FE80::1:1) from 2001:DB8:ACAD:1012::1 (1.1.1.1) Origin IGP,
metric 0, localpref 100, valid, external, best
rx pathid: 0, tx pathid: 0x0
```

```
R2#show bgp ipv4 unicast neighbors 10.1.2.1 advertised-routes
BGP table version is 7, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal
        <output omitted>
        Network          Next Hop       Metric  LocPrf  Weight   Path
*>  192.168.1.0/27    10.1.2.1         0        0      1000      i
*>  192.168.1.64/26   10.1.2.1         0        0      1000      i
*>  192.168.2.0/27    0.0.0.0          0               32768     i
*>  192.168.2.64/26   0.0.0.0          0               32768     i
*>  192.168.3.0/27    10.2.3.3         0        0       300      i
*>  192.168.3.64/26   10.2.3.3         0        0       300      i

Total number of prefixes 6
```

Step 4: Verifying the IP routing tables for IPv4 and IPv6.

```
R2#show ip route bgp | begin Gateway
    Gateway of last resort is not set


        192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
    B   192.168.1.0/27 [20/0] via 10.1.2.1, 00:14:45
    B   192.168.1.64/26 [20/0] via 10.1.2.1, 00:14:16
        192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
    B   192.168.3.0/27 [20/0] via 10.2.3.3, 00:12:13
```

```
 B   192.168.3.64/26 [20/0] via 10.2.3.3, 00:12:13
```

```
R2#show ipv6 route bgp | section 2001
 B   2001:DB8:ACAD:1000::/64 [20/0] via FE80::1:1, Ethernet0/0
 B   2001:DB8:ACAD:1001::/64 [20/0] via FE80::1:1, Ethernet0/0
 B   2001:DB8:ACAD:3000::/64 [20/0] via FE80::3:1, Ethernet0/1
 B   2001:DB8:ACAD:3001::/64 [20/0] via FE80::3:1, Ethernet0/1
```

Task 4: Configure and Verify IPv6 Route Summarization

a. Verify R2 and R3 are receiving 2001:db8:acad:1000::/64 and 2001:db8:acad:1001::/64 from R1.

```
R2# show ipv6 route bgp | section 2001
B   2001:DB8:ACAD:1000::/64 [20/0] via FE80::1:1, Ethernet0/0
B   2001:DB8:ACAD:1001::/64 [20/0] via FE80::1:1, Ethernet0/0
B   2001:DB8:ACAD:3000::/64 [20/0] via FE80::3:1, Ethernet0/1
B   2001:DB8:ACAD:3001::/64 [20/0] via FE80::3:1, Ethernet0/1
```

```
R3# show ipv6 route bgp | section 2001
B   2001:DB8:ACAD:1000::/64 [20/0] via FE80::1:2, Serial1/0
B   2001:DB8:ACAD:1001::/64 [20/0] via FE80::1:2, Serial1/0
B   2001:DB8:ACAD:2000::/64 [20/0] via FE80::2:2, Ethernet0/0
B   2001:DB8:ACAD:2001::/64 [20/0] via FE80::2:2, Ethernet0/0
```

b. Although AS 1000 only has two IPv6 prefixes - 2001:db8:acad:1000::/64 and 2001:db8:acad:1001::/64, this customer has been allocated the entire 2001:db8:acad:1000::/52 prefix (2001:db8:acad:1xxx).

   R1 is configured using the **aggregate-address** command in IPv6 AF mode to summarize its IPv6 prefixes. This is known as a summary route or aggregate route. The **summary-only** option suppresses the more specific prefixes from also being advertised.

```
R1(config)# router bgp 1000
R1(config-router)# address-family ipv6 unicast
R1(config-router-af)# aggregate-address 2001:db8:acad:1000::/52 summary-only
```

c. Verify that R2 and R3 are now receiving the aggregate route and installing it in the IPv6 BGP table.

```
R2# show bgp ipv6 unicast | begin Network
        Network                    Next Hop            Metric  LocPrf  Weight  Path
*  2001:DB8:ACAD:1000::/52  2001:DB8:ACAD:1023::3    0      300      1000      i
*>                          2001:DB8:ACAD:1012::1    0      0     1000   i
<output omitted>
```

```
R3# show bgp ipv6 unicast | begin Network
      Network                    Next Hop              Metric LocPrf Weight Path
*  2001:DB8:ACAD:1000::/52  2001:DB8:ACAD:1023::2   0     500    1000   i
*                           2001:DB8:ACAD:1014::1   0     0     1000   i
<output omitted>
```

d. Verify that R2 and R3 are now receiving the aggregate route and it is installed in the IPv6 routing table.

```
R2# show ipv6 route bgp | section 2001
B   2001:DB8:ACAD:1000::/52 [20/0] via FE80::1:1, Ethernet0/0
B   2001:DB8:ACAD:3000::/64 [20/0] via FE80::3:1, Ethernet0/1
B   2001:DB8:ACAD:3001::/64 [20/0] via FE80::3:1, Ethernet0/1

R3# show ipv6 route bgp | section 2001
B   2001:DB8:ACAD:1000::/52 [20/0] via FE80::1:2, Serial1/0
B   2001:DB8:ACAD:2000::/64 [20/0] via FE80::2:2, Ethernet0/0
B   2001:DB8:ACAD:2001::/64 [20/0] via FE80::2:2, Ethernet0/0
```